

internal-external rotation 90 degree-abducted — random spline (ns df=4, D diagonal), trimmed to overlap

Sternoclavicular — Protraction(+)/retraction(-)

not modelled

Sternoclavicular — Depression(+)/elevation(-)

not modelled

Sternoclavicular — Backwards(+)/forward(-) rot

not modelled

Acromioclavicular — Protraction(+)/retraction(-)

not modelled

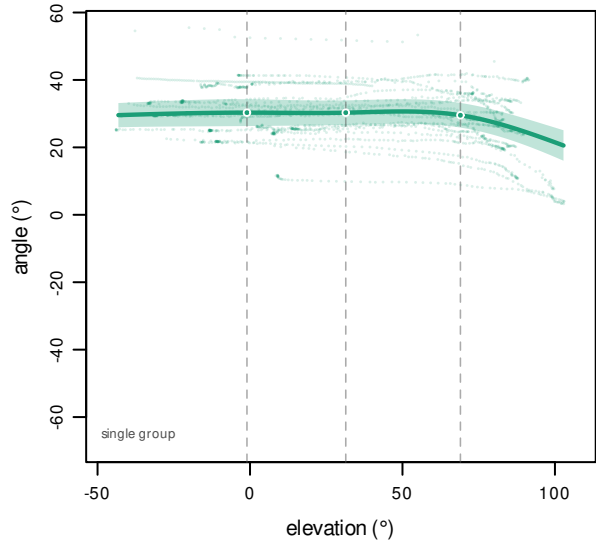
Acromioclavicular — Medial(+)/lateral(-) rot

not modelled

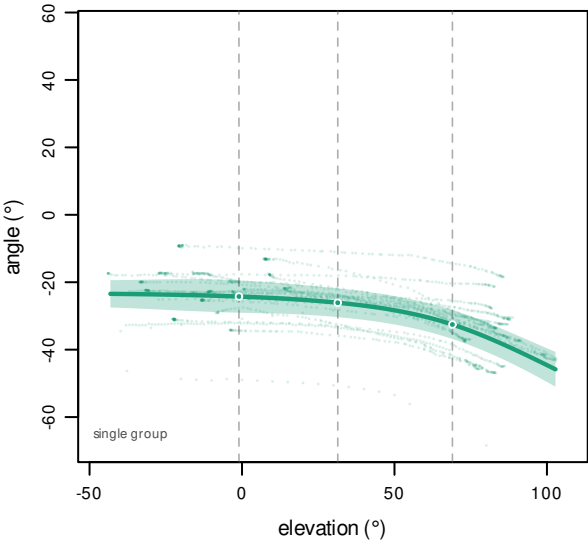
Acromioclavicular — Posterior(+)/anterior(-) tilt

not modelled

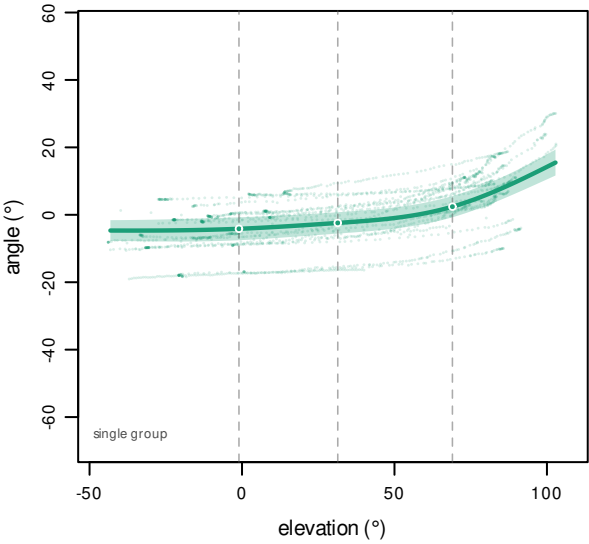
Scapulothoracic — Protraction(+)/retraction(-)



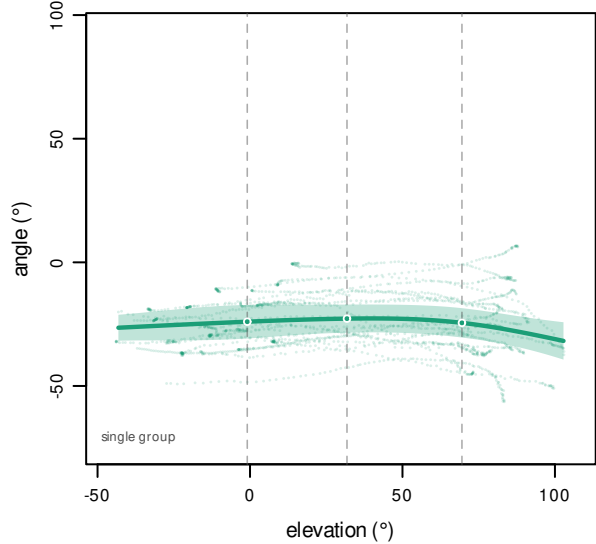
Scapulothoracic — Medial(+)/lateral(-) rot



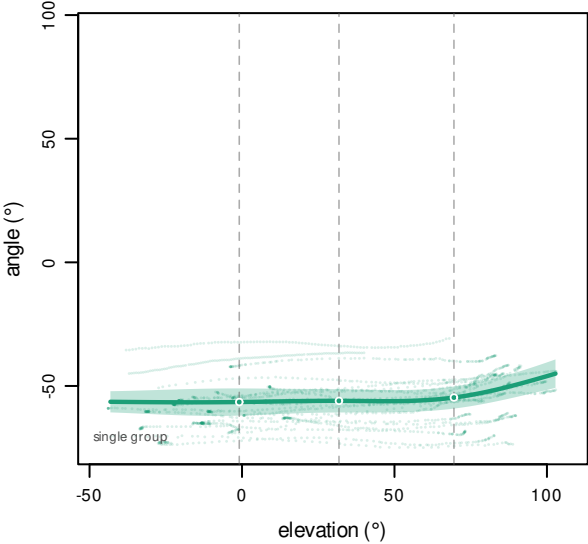
Scapulothoracic — Posterior(+)/anterior(-) tilt



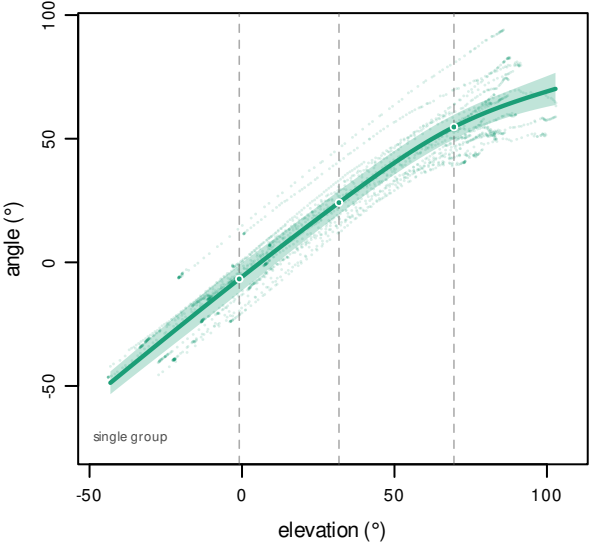
Glenohumeral — Plane of elevation



Glenohumeral — Elevation(-)/Depression(+)



Glenohumeral — Int(+)/ext(-) rotation



orange = ex vivo, green = in vivo · line = population curve, band = 95% CI · dashed verticals = interior knots, dots = fitted value at each knot

op-right: one dot per in-vivo difference coefficient, read LEFT TO RIGHT as gamma_0, gamma_1, ... — the index is printed under each dot. gamma_0 = constant level shift (condin vivo); gamma_k = ns_k:condin vivo, a shape term
filled = that coefficient passed |coef/se| >= 1.96, hollow = did not. stars = JOINT test on ALL difference coefficients at once, BH-FDR: *** p<0.001, ** p<0.01, * p<0.05, ns = not significant

bottom-left: shoulders ex/in · studies ex/in | Wald ref/diff counts | mean and max |Δ| in degrees. Curves are drawn only where ≥2 shoulders have data.